

Time-Reversal CNN-Based S-NOFDM Scheme for Underwater Acoustic Communication

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Abstract—Multicarrier modulation shows the advantages of improving bandwidth efficiency in underwater acoustic (UWA) communications, whereas the more spectrum-efficient nonorthogonal frequency division multiplexing (NOFDM) is seldom investigated in the underwater scenario. The main idea of the NOFDM is to use overlapped subcarriers for increasing spectral efficiency. However, the NOFDM suffers from severe distortions both in the time and frequency domains because of the UWA channel's multipath propagation and Doppler spread. Considering these problems, we propose a novel time-reversal convolutional neural network (TR-CNN) based sparse NOFDM (S-NOFDM) scheme for UWA communication. The S-NOFDM waveform is determined by solving the sparse representation of input symbols under a set of nonorthogonal subcarriers, which can allocate symbols sparser to subcarriers and reduce distortion caused by multipath propagation and Doppler spread. The proposed TR-CNN is constructed with a time-reversal (T-R) convolutional layer where the T-R process is used to deal with multipath propagation. After the received signal has passed through the T-R layer, the main path of the UWA channel strength is enhanced and the other paths are weakened. The TR-CNN is also used to learn the inherent information between S-NOFDM symbols, improving UWA channel adaptability. After the training, the proposed scheme can be applied in the real-time UWA communication implementation. The simulation and experimental results prove that the proposed scheme outperforms the traditional scheme in signal detection and recovery. It is also robust with general capabilities in various UWA channel environments.

Index Terms—Signal detection performance, sparse nonorthogonal frequency division multiplexing (S-NOFDM), time-reversal convolutional neural network (TR-CNN), underwater acoustic (UWA) communication.

I. INTRODUCTION

THE challenging underwater acoustic (UWA) channel increases the difficulties to achieve efficient and reliable UWA communication. With limited bandwidth and severe multipath, the communication in UWA faces more complex channel conditions [1], [2], causing signal detection and recovery quite problematic.

To deal with limited bandwidth, in recent years, orthogonal frequency division multiplexing (OFDM) has been broadly

investigated to improve bandwidth efficiency. OFDM is based on the orthogonal subcarriers, which are resistant to intersymbol interference (ISI) via cyclic prefix (CP), and can achieve reliable high-rate transmission in the UWA channel [3]. While OFDM is sensitive to Doppler shifts, the CP-related techniques for removing inter-carrier interference (ICI)/ISI cost extra spectrum resources, which is not welcome in UWA channels.

To further improve spectral efficiency, some research has focused on nonorthogonal frequency division multiplexing (NOFDM) modulation schemes. The NOFDM scheme features nonorthogonality/overlapping of symbols/subcarriers as shown in [4], [5], and [6], essentially it exploits the ISI/ICI to improve the bandwidth efficiency. The technique of a spectrally efficient frequency division multiplexing system is proposed for wireless communication in [7], [8], and [9], where the spectral efficiency is improved by compressing the subcarrier spacing.

However, the overlapping of subcarriers in NOFDM will introduce more difficulties in processing signals passing the UWA channel. There is time dispersion caused by multipath propagation and frequency dispersion due to the Doppler effect. These factors can aggravate ISI/ICI in NOFDM systems, make signal detection more challenging, and degraded signal detection performance [10], [11]. Thus, complex receiver processing is needed for dealing with the harsh UWA channel [10], [12], [13].

From the aforementioned research, it can be read that the performance of UWA communication is greatly related to the UWA channel conditions. In some previous work, methods of reducing the overlapped subcarrier interference problem in NOFDM schemes have been investigated. A sparse NOFDM (S-NOFDM) modulation technique by duplicating and overlapping subcarriers is investigated in [14]. The S-NOFDM waveform is determined by solving the sparse representation of input symbols under a set of nonorthogonal subcarriers. The sparse representation of symbols uses fewer coefficients in representing signals and distributes symbols more separately among subcarriers so that spectral resources are exploited more efficiently. The improved bit error rate (BER) and spectral efficiency are very attractive for bandwidth-limited UWA communication systems [14]. However, the existing S-NOFDM only focuses on improving spectral efficiency and cannot handle the complex UWA channel effectively. The S-NOFDM symbols experienced multipath propagation and Doppler spread where time and frequency dispersion occur, causing severe ISI and ICI.

In this article, considering the complicated UWA channel, we propose a new S-NOFDM-based scheme to reduce the interference caused by multipath propagation and the Doppler

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spread. To deal with the multipath effect, the time-reversal (T-R) process is used in underwater communication to deal with harsh underwater channel features [15], [16]. It is an effective method to mitigate the long delay spread by the multipath effect of the UWA channel and has been widely used in OFDM communications [12], [13]. However, T-R can only deal with multipath propagation, other UWA channel problems such as Doppler spread and noises cannot be solved. Recently, the state-of-the-art convolutional neural network (CNN) technique is utilized in multicarrier communication schemes. It proves to be highly effective in dealing with distortion caused by Doppler spread and noises in the UWA channel. By using CNN for interfered signal demodulation, BER improvements are obtained due to its powerful learning capabilities [17], [18], [19], [20]. However, if only received signals are used for training, CNN cannot manage different channel environments.

Focusing on the above-mentioned issues, we constructed a time-reversal CNN (TR-CNN) and applied it to the S-NOFDM UWA communication system. We noticed that the T-R process can be considered as a matched filter for the channel impulse response (CIR) and the main idea is to reverse and convolve the signal. This idea is also adopted by CNN. So, we modified the CNN network and one convolutional layer performing the T-R process is responsible for dealing with the multipath propagation. The constructed TR-CNN can also handle other S-NOFDM symbol distortions when transmitting in the UWA channel and improve signal detection and recovery performance. The main contributions are listed as follows.

- 1) We first apply the S-NOFDM scheme in the underwater scenario and improve its adaptability to the UWA channels. The S-NOFDM allocates symbols sparser to subcarriers, which can lighten the UWA channel distortion caused by multipath propagation and Doppler spread.
- 2) The TR-CNN is constructed to fully exploit the channel state information and the inherent information between S-NOFDM symbols. The proposed TR-CNN is constructed with a T-R convolutional layer where the estimated CIR is processed to enhance the main path strength. The real and imaginary parts of the nonorthogonal subcarriers are parallelized to keep S-NOFDM signal inherent information. Multipath information and S-NOFDM symbols are trained to improve the SNR at the receiver and allow the newly constructed CNN to learn the feature information between nonorthogonal subcarriers. As a result, the proposed TR-CNN-based S-NOFDM scheme improves the generalization ability and robustness to various channel environments.

Through simulation and experiment, the results effectively prove that the proposed scheme outperforms the traditional signal detection method, and the network is robust to various channel environments, CP length, and the number of pilots.

The rest of this article is organized as follows. In Section II, the system model of the proposed scheme is explained. In Section III, the numerical model and scheme details including the T-R process implemented by TR-CNN, the channel estimator, the structure of the TR-CNN, and the training procedure are presented. In Section IV, the simulation and experimental results

are demonstrated and analyzed. Finally, Section V concludes this article.

II. SYSTEM MODEL

In this section, the system model of the proposed schemes is explained, and we focus on describing the workflow and numerical theory of the proposed scheme. Fig. 1 depicts the proposed TR-CNN-based S-NOFDM system. The transmitter in the system handles passing binary inputs to a mapper, converting mapped signals through the S-NOFDM modulator to the time domain, adding CPs, and finally sending modulated sequences to the UWA channel. On the receiver side, the CP of the received signal was removed first. Then, the signal is processed with the T-R process and fed into CNN together with the channel factor for demodulation.

A. Transmitter

At the transmitter, the randomly generated binary data sequence $\mathbf{b}_s$ is first mapped by quadrature amplitude modulation. Then, the mapped signal $\mathbf{d}_s$ is converted from serial to parallel to obtain $\mathbf{d} \in \mathbb{C}^N$, which is the input signal of the S-NOFDM modulator.

The S-NOFDM modulator is designed and adopted with the sparse nonorthogonal scheme in [14]. A new subcarrier matrix $\mathbf{G}$ is generated by extending the subcarrier from an orthogonal Fourier basis Φ ($\Phi \in \mathbb{C}^{K \times K}$) for modulation. Each selected OFDM subcarrier is shifted into a group of overlapping subcarriers with different time delays.

The structure of matrix $\mathbf{G}$ is illustrated in Fig. 2. The matrix $\mathbf{G}$ is the subcarrier matrix that contains all the subcarriers as its columns. In our scheme, K subcarriers are each extended with M times, with $(M-1) \cdot \Delta$ time shifts. So, $\mathbf{G} \in \mathbb{C}^{N \times R}$, $N = (M-1) \cdot \Delta$ and $R = M \cdot K$. The total number of subcarriers is R ; ($R = K \cdot M$), where M is the number of expanding subcarriers at the same frequency and Δ is the length of the time-shifting unit.

We denote a set of complex-valued mapped symbols as $\mathbf{d} \in \mathbb{C}^{N \times 1}$. So, the process of finding an optimal representation with a small number of significant coefficients that suits certain restrictions is termed sparse coding. We consider the S-NOFDM as representing signal $\mathbf{d}$ with sparse coefficients. The modulation of the signal can be described as

$$\mathbf{d} = \mathbf{G}\mathbf{x} \quad (1)$$

where $\mathbf{x} \in \mathbb{C}^R$ is the sparse coefficient vector under the matrix $\mathbf{G}$, showing the sparsest R ($R > N$) subcarriers to represent the modulated signal $\mathbf{x}$ with R symbols.

The modulated signal $\mathbf{x}$ is produced by solving the sparse representation of the input signal $\mathbf{d}$ under the matrix $\mathbf{G}$. Using l_1 -norm as the sparse prior, S-NOFDM modulation is equivalent to the following l_1 -minimization problem:

$$\min \|\mathbf{x}\|_1 \text{ subject to } \mathbf{d} = \mathbf{G}\mathbf{x}. \quad (2)$$

Then, the obtained sparse solution $\mathbf{x}$ is passed to the parallel to serial for further processing. Subsequently, CP insertion is performed to obtain the transmitted signal $\tilde{\mathbf{x}}_s$. After through the

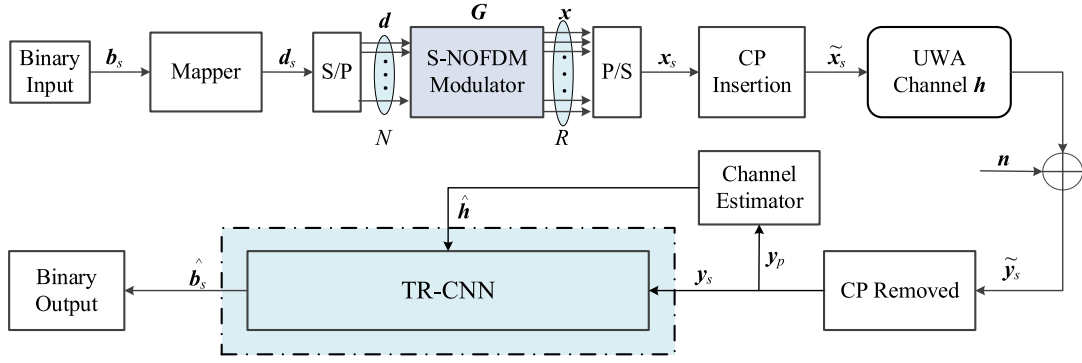


Fig. 1. System model of TR-CNN-based S-NOFDM UWA communication scheme.

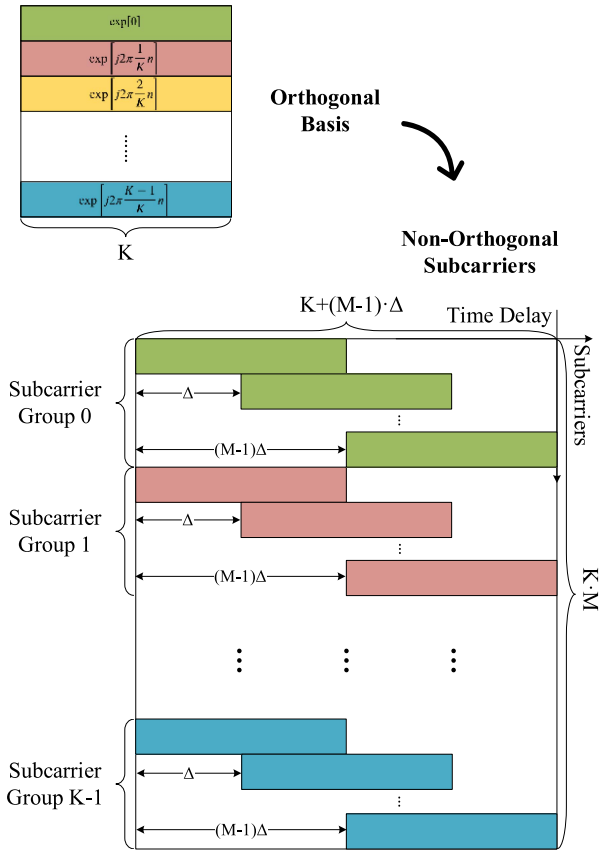


Fig. 2. Illustration of S-NOFDM matrix generation.

UWA channel, the received signal $\tilde{y}_s$ is given by

$$\tilde{\mathbf{Y}}_s(t) = \tilde{\mathbf{x}}_s(t) \otimes \mathbf{h}(t) + \mathbf{n}(t) \quad (3)$$

where the operator $\otimes$ corresponds to the convolution, and $h(t)$ and $n(t)$ are the time domain channel response and additive white Gaussian noise (AWGN) with zero-mean, respectively.

B. UWA Channel

The UWA channel is different from the terrestrial channel and is characterized by fast time-varying, long delay spread, and Doppler shifts. We generate the channel using a model based on

statistical properties proposed by Qarabaqi and Stojanovic [21], which consider the physical properties of acoustic propagation as well as the effects of random variations. The effects of multipath and Doppler effects are modeled by the transfer function as follows:

$$\mathbf{H}(f, t) = \bar{\mathbf{H}}_0 \sum_p \mathbf{h}_p(t) \tilde{\gamma}_p(f, t) e^{-j2\pi f \tau_p} \quad (4)$$

where $\bar{\mathbf{H}}_0$ is the amplitude matrix with filter effects for all the paths and p is the number of paths. $\mathbf{h}_p(t)$ and $\tau_p(t) = \tau_{p0} - a_p(t)$ are the path gain coefficient and delays of the p th path that vary in time according to the random Doppler scaling factors a_p and $\tilde{\gamma}_p(f, t) = \gamma_p(f, t) e^{j2\pi a_p f t}$, $p = 1, 2, \dots$, the scattering coefficients are modeled as complex-valued Gaussian processes. $\mathbf{H}(f, t)$ is a matrix that represents the time-varying channel transfer function in the frequency domain. It is denoted by $\mathbf{h}(f, t)$ in the time domain.

C. Receiver

In a traditional S-NOFDM receiver, the received signal $\tilde{y}_s$ is removed from the CP and a series of parallel conversions are performed to obtain the distorted waveforms $\mathbf{y}$. Then, the latent waveform $\hat{\mathbf{x}}$ is recovered by time-domain channel equalization. The S-NOFDM demodulator corresponding to the modulator is used to recover the demodulated signal $\hat{\mathbf{d}}$ through $\hat{\mathbf{x}}$

$$\hat{\mathbf{D}} = \mathbf{G}\hat{\mathbf{x}}. \quad (5)$$

Finally, the recovered data $\hat{\mathbf{d}}$ are processed for parallel to serial conversion and remapped to obtain the binary output $\hat{b}_s$.

Compared with the OFDM modulation scheme, the number of different subcarrier frequencies in the modulated dictionary $\mathbf{G}$ is less than that in the reference OFDM system leading to higher spectral efficiency in S-NOFDM systems, but it suffers from heavy ISI/ICI and UWA channel deterioration.

In our proposed TR-CNN-based S-NOFDM receiver, the convolution operation of TR-CNN is used to implement the T-R process and extract signal features that improve signal detection performance and demodulation reliability. We first use the channel estimator to obtain the estimated CIR as a part of the TR-CNN input. The estimated CIR $\hat{\mathbf{h}}$ and the received CP removal signal $\mathbf{y}_s$ are the inputs to the T-R process. After

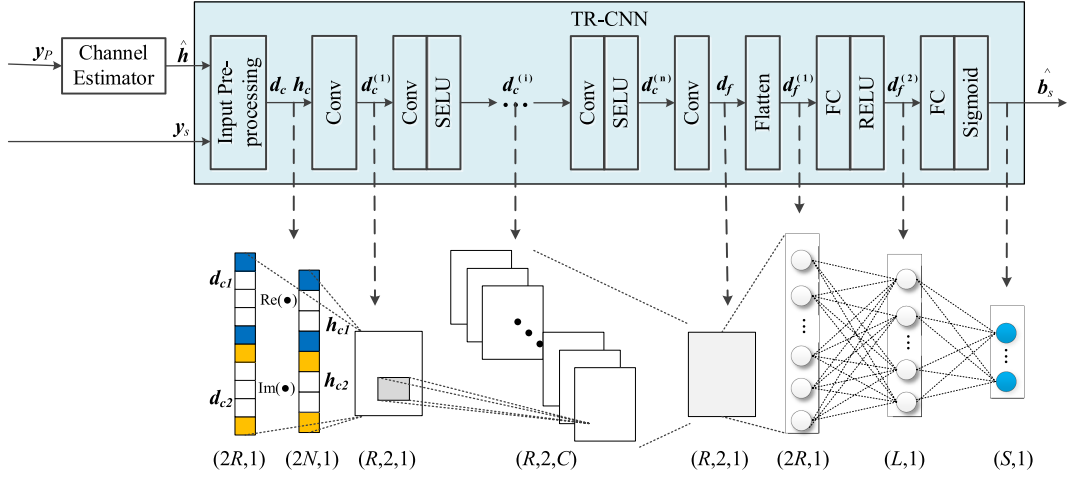


Fig. 3. Detailed architecture of proposed TR-CNN process for UWA communication systems.

enhancing the main path signal energy, they are fed to the various neural network layers for extracting the inherent information features between nonorthogonal subcarriers. Passing through the CNN, the binary bits are finally demodulated for output. The detailed architecture of the proposed TR-CNN-based S-NOFDM receiver that replaces the traditional S-NOFDM receiver is shown in Fig. 3, which includes the channel estimator and the TR-CNN.

In this article, we use the block-based least square (LS) estimator at the receiver and denote the received pilot signal by $\mathbf{y}_P \in \mathbb{C}^{N \times 1}$. The frequency-domain signal $\mathbf{Y}_P(k)$ is obtained by fast Fourier transform (FFT) and then LS channel estimation is performed together with the transmitter pilot $\mathbf{x}_P(k)$

$$\begin{aligned} \hat{\mathbf{H}}_{\text{LS}}(k) &= [\mathbf{x}_P^H(k) \mathbf{x}_P(k)]^{-1} \mathbf{x}_P^H(k) \mathbf{Y}_P(k) \\ &= \mathbf{x}_P^{-1}(k) \mathbf{Y}_P(k) \end{aligned} \quad (6)$$

$$\hat{\mathbf{H}}(t) = \sum_{k=0}^{N-1} \hat{\mathbf{H}}_{\text{LS}}(k) e^{\frac{j2\pi kt}{N}} \quad (7)$$

where the superscript $(\cdot)^H$ indicates the Hermitian transpose, and $\mathbf{x}_P(k)$ and $\mathbf{Y}_P(k)$ are the pilot symbol carried the k th subcarrier and the corresponding receive symbol, respectively. Vector $\hat{\mathbf{h}}(t) \in \mathbb{C}^{N \times 1}$ is the estimated CIR obtained by $\hat{\mathbf{H}}_{\text{LS}}(k)$ via inverse FFT.

The TR-CNN consists of input preprocessing, several convolutional (Conv) layers, one flatten layer, and two fully connected (FC) layers. The functions and parameters of these layers will be explained in the following section.

III. TIME-REVERSAL (T-R) AND CNN IMPLEMENTATION

The T-R process is also a convolution operation that can be implemented into the CNNs jointly. In this section, we will demonstrate the TR-CNN implementations.

A. T-R Process Implemented by TR-CNN

CNNs are simple neural networks that use convolution in at least one of their layers instead of the general matrix multiplication, and it is a convolutional layer. The convolution layer extract features by regularly scanning the input data $\mathbf{I}$ via convolution kernel $\mathbf{K}$. The output of a 1-D convolution is represented as follows [17]:

$$\mathbf{Z}(t) = [\mathbf{K} \otimes \mathbf{I}](t) = \sum_m \mathbf{I}(t+m) \mathbf{K}(m) \quad (8)$$

where m denotes the index of the input data $\mathbf{I}$. The convolution operation in the convolution layer is the related function of cross-correlation, which is the same as convolution but without flipping the convolution kernel $\mathbf{K}$. In the S-NOFDM UWA communication system, the T-R processing of the received signal is expressed as follows:

$$\begin{aligned} \mathbf{S}'(t) &= \mathbf{y}_s(t) \otimes \mathbf{h}'(t) \\ &= \mathbf{h}'(-t) \otimes \mathbf{y}_s(t) \\ &= \sum_p \mathbf{h}'(-p) \mathbf{y}_s(t-p) \\ &= \sum_p \mathbf{h}'(p) \mathbf{y}_s(t+p) \end{aligned} \quad (9)$$

where $\mathbf{y}_s(-t)$ is the T-R of the received signal $\mathbf{y}_s(t)$, $\mathbf{h}'(t)$ is the estimated CIR obtained using the channel estimation algorithm, and p is the number of multipath in the UWA channel. The main purpose of the T-R process is to obtain the reversal and convolution of the signal. While the convolution in CNN is precise without flipping the convolution kernel, we, therefore, utilize TR-CNN's convolution operation to implement T-R signal processing, which not only reduces the calculation complexity without prior signal reversal but also improves the SNR performance at the receiver.

B. Structure of the TR-CNN

1) *Input Preprocessing*: Different from the deep learning-based multicarrier receivers [18], [22], [23], and [24], the proposed network is not only fed with received signals. Since the distorted received signal contains the ISI/ICI effects of the current channel environment, if only fed into the network, the network learns the inherent information between symbols, which is dependent on the current channel environment. For fast time-varying UWA channels, the network parameters must be updated to track the channel changes, which significantly increases the training overhead. To increase the generalization capability and robustness of the network to the environment, we need to obtain the estimated CIR and use it as part of the TR-CNN input to make the network adaptable to a fast time-varying UWA channel.

Therefore, the TR-CNN treats both the estimated CIR $\hat{\mathbf{h}} \in \mathbb{C}^{N \times 1}$ and the received information signal $\mathbf{y}_s \in \mathbb{C}^{R \times 1}$ as training data inputs and outputs a vector of predicted binary data $\hat{\mathbf{b}}_s$. First, the complex vectors $\mathbf{y}_s$ and $\hat{\mathbf{h}}$ need to be transformed by concatenating the real and the imaginary parts into a column of real numbers before they are fed into the following neural network layers, which are represented as $\mathbf{d}_c \in \mathbb{R}^{2R \times 1}$, $\mathbf{h}_c \in \mathbb{R}^{2N \times 1}$ in Fig. 3:

$$\mathbf{d}_c = [\mathbf{d}_{c1}, \mathbf{d}_{c2}] = [\text{Re}(\mathbf{y}_s), \text{Im}(\mathbf{y}_s)] \quad (10)$$

$$\mathbf{H}_c = [\mathbf{h}_{c1}, \mathbf{h}_{c2}] = [\text{Re}(\hat{\mathbf{h}}), \text{Im}(\hat{\mathbf{h}})] \quad (11)$$

where $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ denote the real and imaginary parts, respectively.

2) *Conv Layers*: The input vectors $\mathbf{d}_c$ and $\mathbf{h}_c$ are convolved through TR-CNN to implement the T-R process and the output $\mathbf{d}_c^{(1)}$ is obtained by convolving $\mathbf{d}_{c1}$, $\mathbf{d}_{c2}$, $\mathbf{h}_{c1}$, and $\mathbf{h}_{c2}$, respectively, with the equations denoted as (12) and (13). They represent the real and imaginary parts of the complex vectors $\mathbf{y}_s$ and $\hat{\mathbf{h}}$, respectively. Separating the real and imaginary parts can make full use of the multipath information compared to using only the real part, or the amplitude. In this process, $\mathbf{h}_{c1}$ and $\mathbf{h}_{c2}$ function as convolutional kernels and the convolution process utilizes the UWA channel multipath information component to mitigate the long delay spread and improve the received SNR. This mitigation of the delay spread can be seen in the analysis of the next section, which concentrates most of the energy in the main path and weakens the distortion of the received signal from the other paths.

$$\mathbf{d}_c^{(1)}(i) = \sum_{n=0}^{N-1} \mathbf{h}_{c1}(n) \mathbf{d}_{c1}(i-n) - \mathbf{h}_{c2}(i) \mathbf{d}_{c2}(i-n) \quad (12)$$

$$\mathbf{d}_c^{(1)}(j) = \sum_{n=0}^{N-1} \mathbf{h}_{c1}(n) \mathbf{d}_{c2}(j-n) + \mathbf{h}_{c2}(i) \mathbf{d}_{c1}(j-n). \quad (13)$$

Furthermore, we will parallel $\mathbf{d}_c^{(1)}(i)$ and $\mathbf{d}_c^{(1)}(j)$ to form 2-D matrices $\mathbf{d}_c^{(1)} = [\mathbf{d}_c^{(1)}(i), \mathbf{d}_c^{(1)}(j)] \in \mathbb{R}^{R \times 2 \times 1}$ (single-channel

input), which then is fed into the convolutional layer. At this stage, the input vector $\mathbf{d}_c^{(1)}$ has removed some of the distortions from the UWA channel, so that the convolutional layer learns the inherent information between the S-NOFDM symbols, regardless of the current channel environment. The convolutional layer has fewer training parameters and can learn more complex information features, which is more suitable for training S-NOFDM symbols compared to DNN. In the convolutional layer, the convolutional kernel k is utilized to extract features from the local neighborhood on feature maps of the previous layer. Then, an additive bias is applied, and the activation function is the scaled exponential linear unit activation function

$$f_{\text{selu}}(x) = \lambda \begin{cases} x & \text{if } x > 0 \\ \alpha e^x - \alpha & \text{if } x \leq 0 \end{cases} \quad (14)$$

which has self-normalization characteristics and its variance stabilization also avoids gradient explosion and gradient disappearance [25]. We denote the output of the i th layer as $\mathbf{d}_c^{(i)} \in \mathbb{R}^{R \times 2 \times C}$ (C is the output channels), formally, the unit value of the position $[x, y]$ in the j th feature map is expressed as follows:

$$d_{cj}^{(i)}[x, y] = f_{\text{selu}} \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} d_c^{(i-1)}[x+p, y+q] k_j^{(i)}[p, q] + b_j^{(i)} \quad (15)$$

where $b_j^{(i)}$ is the bias for this feature map, $k_j^{(i)}[p, q]$ is the value at the position $[p, q]$ of the kernel connected to the j th feature map, and P and Q are the height and width of the kernel, respectively.

Notably, we design n convolutional layers, the n th layer has no activation function and the output channel is set to 1, so the output of finally the Conv Layers is $\mathbf{d}_f \in \mathbb{R}^{R \times 2 \times 1}$, as shown in Fig. 3.

3) *Flatten Layer and FC Layers*: Last, we use one flatten layer and two FC layers. A flatten layer transforms the 2-D matrix $\mathbf{d}_f$ into a 1-D vector $\mathbf{d}_f^{(1)} \in \mathbb{R}^{2R \times 1}$. After flattening, the vector $\mathbf{d}_f^{(1)}$ is fed into two FC layers. The first and the second FC layer use the rectified linear unit (RELU) activation function $f_{\text{relu}}(x) = \max(0, x)$ and Sigmoid activation function $f_{\text{sigmoid}}(x) = \frac{1}{1+e^{-x}}$, respectively. Dropout is added to the first FC layer to avoid overfitting. The output of the FC Layers can be expressed as

$$\mathbf{d}_f^{(2)} = f_{\text{relu}}(\mathbf{w}^{(1)} \mathbf{d}_f^{(1)} + \mathbf{c}^{(1)}) \quad (16)$$

$$\hat{\mathbf{b}}_s = f_{\text{sigmoid}}(\mathbf{w}^{(2)} \mathbf{d}_f^{(2)} + \mathbf{c}^{(2)}) \quad (17)$$

where $\mathbf{w}^{(1)} \in \mathbb{R}^{L \times 2R}$ and $\mathbf{w}^{(2)} \in \mathbb{R}^{S \times L}$ represent the training weights of the first and the second FC layer, respectively. $\mathbf{c}^{(1)} \in \mathbb{R}^{L \times 1}$ and $\mathbf{c}^{(2)} \in \mathbb{R}^{S \times 1}$ represent the training bias of the first and the second FC layer, respectively. $\mathbf{d}_f^{(2)} \in \mathbb{R}^{L \times 1}$ and $\hat{\mathbf{b}}_s \in \mathbb{R}^{S \times 1}$ is the output of the first and the second FC layer, respectively.

C. Training Procedure

The proposed TR-CNN-based S-NOFDM receiver includes training and testing stages. In the training stage, we first generate random data sequences as the transmitted symbols and fix the

pilot symbols to be used for channel estimation. Then, the current random training channel is simulated based on the statistical properties of the UWA channel [21]. The received S-NOFDM signal is obtained by undergoing the current channel distortion (including channel noise), the distorted received signal, and the estimated CIR collected as the training data, and the originally transmitted data are collected as the training labels. The amount of training set and test set ratio is 4:1. The network parameters of the TR-CNN include convolutional and FC layers; with the iteration period increase, the optimizer continuously updates these parameters, and the loss function converges accordingly. In this article, we use the mean square error (MSE) loss function to minimize the difference between the output of the TR-CNN $\hat{\mathbf{b}}_s$ and the training labels $\mathbf{b}_s$, which can be expressed as

$$\text{Loss} = \frac{1}{V} \sum_{v=0}^{V-1} \|\hat{\mathbf{b}}_s(v) - \mathbf{b}_s(v)\|^2 \quad (18)$$

where $v = 0, 1, \dots, V-1$, V is the batch size. When the loss function reaches the preset threshold, the training parameters of TR-CNN will stop updating and the training stage is finished. In the testing stage, the pretrained parameters of the TR-CNN can be used directly at the receiver to perform real-time detection for the received signals. Once the distorted received signal $\mathbf{y}_s$ and the estimated CIR $\hat{\mathbf{h}}$ are fed into the pretrained TR-CNN, it can estimate the bits immediately according to the structure described in Section III-B.

D. Computational Complexity Analysis

The complexity of the proposed scheme can be separately discussed by the modulator and the receiver. $O(1)$ is assumed as the computational complexity for one multiplication.

In the modulation process, S-NOFDM modulation is solving the problem in (2). By utilizing the BP algorithm, the computational complexity of the modulator is $O(N^3)$ with the data size of N . Or if a pseudoinverse method is utilized for calculating x in (2), the complexity is $O(N^2)$.

In the traditional receiver, bits recovery is performed with channel estimation, equalization, and symbol demodulation. If using the LS for channel estimation, the complexity is $O(N^2)$. If orthogonal matching pursuit (OMP) is adopted, the computational complexity is $O(pN)$, where p is the pilot number. In our proposed receiver, the channel equalization and demodulation are processed all together in the TR-CNN network. The overall computational complexity of the Conv Layers is $O(MN)$, where M is the node number and N is the input data length.

IV. SIMULATION AND EXPERIMENTAL RESULTS

In this section, we give the specific parameters of the S-NOFDM system and the UWA channel, the hyperparameters of the TR-CNN. We discuss in detail the proposed scheme in simulation and experimental results, the BER performance based on simulation, and the experimental results. Results proved that the proposed TR-CNN-based S-NOFDM scheme has significant gain compared to the traditional NOFDM scheme and can obtain

TABLE I
CONFIGURATIONS OF S-NOFDM SYSTEM

Parameters	Value
Number of OFDM subcarriers N	1024
Number of selected OFDM subcarriers K	512 / 384
Number of expanding subcarriers M	3
Number of S-NOFDM subcarriers R	1536 / 1152
Number of pilot symbols P	1024
CP length	1/4

TABLE II
CONFIGURATIONS OF THE UWA CHANNEL

Surface height (m)	100	100	100
TX height (m)	50	50	50
RX height (m)	60	60	60
Distance (km)	1.3	1.3	5.0
Frequency Range (kHz)	22-28	22-28	22-28
Maximum delay (ms)	25	25	40
Doppler shifts (Hz)	0	15	30

TABLE III
HYPERPARAMETERS OF THE TR-CNN

Parameters	Value
Number of Conv Layers n	5
Kernel size k	(3, 2)
Number of output channels C	16
Zero padding	same
Stride	(1, 1)
Number of neurons in the first FC layer L	1024
Dropout rate	0.5
Number of neurons in the second FC layer S	16
batch size	64
Optimizer	Adam
Learning rate	10^{-3}

higher spectral efficiency under the UWA channel by reducing the CP length and the number of pilots.

A. Configurations

The S-NOFDM communication system and UWA channel environment are shown in Tables I and II, respectively. The hyperparameters of the TR-CNN are listed in Table III. The training procedure is only performed in Env1 with the received SNR of 20 dB. The test SNR values are in the range of 0–20 dB with a step size of 5 dB. The packet size of each frame is 10 240 b and for each simulation, the random bits of 200 packets are sent. To verify the robustness and generalization capability of the proposed TR-CNN-based S-NOFDM scheme, three different UWA channel environments (Env1–Env3) such as delay spreads and Doppler shifts were simulated and the time-domain CIR is illustrated in Fig. 4. We will evaluate the generalization capability and robustness of the proposed TR-CNN-based S-NOFDM scheme by assessing the various channel environments, different CP lengths, and the number of

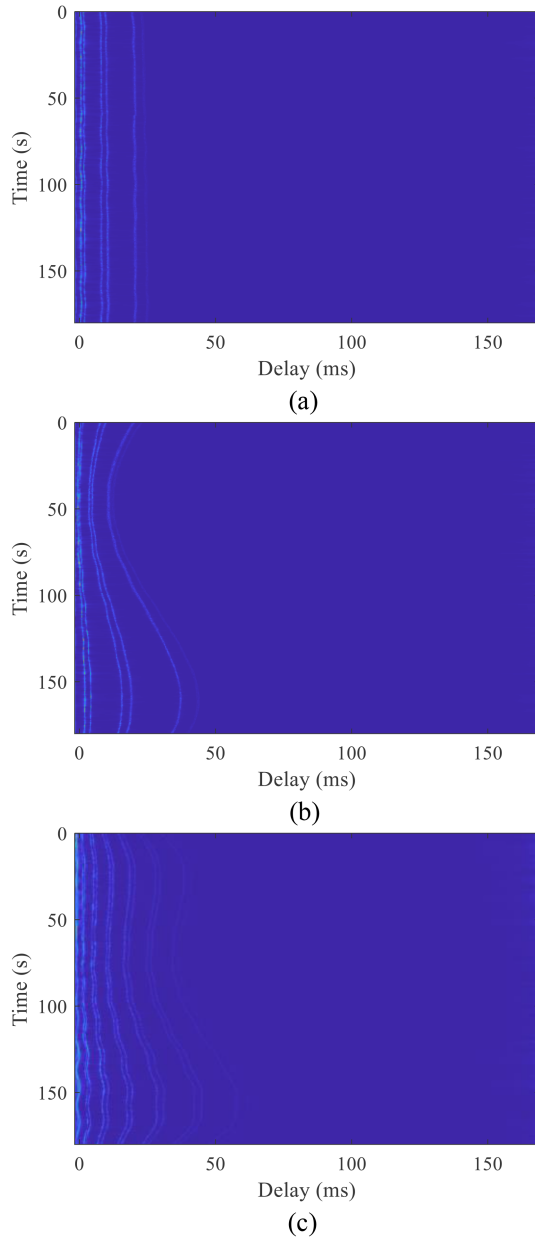


Fig. 4. Time-domain channel impulse response. (a) Env1. (b) Env2. (c) Env3.

pilots. Moreover, we also handle real data and further analyze the robustness under the mismatched channels.

B. System Performance Analysis

To mitigate the long delay spread caused by the multipath effect in the UWA channel, we utilize the multipath information component in TR-CNN to implement the T-R process by convolution operation and describe the specific implementation in Section III-B.

Fig. 5(a) and (b) shows the channel condition of Env1 before the Conv Layers and after the Conv Layers. It can be seen that after convolution of the estimated CIR and distorted received signal through the Conv Layers, the long delay spread is greatly mitigated using the multipath information component while

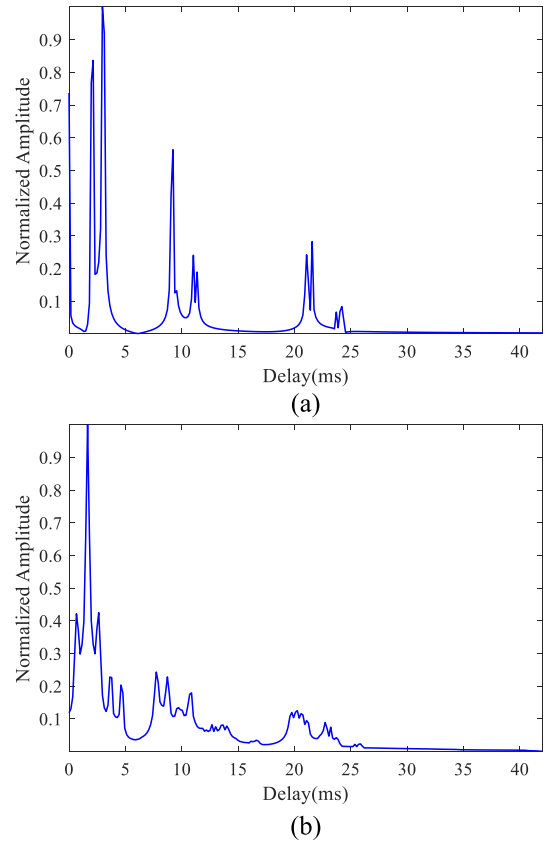


Fig. 5. BER performance comparisons of the proposed TR-CNN-based S-NOFDM scheme with different CP lengths. (a) Channel condition of Env1 before Conv Layers. (b) Channel condition of Env1 after Conv Layers.

enhancing the main path and weakening the other paths. This is necessary for the S-NOFDM UWA communication system to greatly mitigate long delay spread and improve the received SNR.

We then analyze the signal detection performance with different spectral efficiencies in UWA communication systems. The BER performance is compared among the proposed TR-CNN-based S-NOFDM receiver, traditional LS zero-force (ZF) receiver, and the receiver proposed in [10], which utilized resampling, basis expansion model channel modeling, the OMP channel estimation, and MSE time-domain equalizer. To find the optimal performance of the proposed scheme under ideal conditions, i.e., no ICI, multipath, delay spread, and Doppler shift, we perform a simulation in the same AWGN channel. Other parameters in the figure are illustrated in Table I, and the traditional receiver performance of the OFDM system is used as a baseline to compare the spectral efficiency and BER of S-NOFDM.

In Fig. 6(a), the dashed line of OFDM and S-NOFDM showed the theoretical BER result. The performance of S-NOFDM is slightly better than that by the OFDM schemes owing to the sparse representation of signals in the transmitter [14], [26].

When comparing different receiver schemes, we can see that the results of the receiver proposed in [10] (OMP-MSE) are better than the LS-ZF because the OMP algorithm utilized

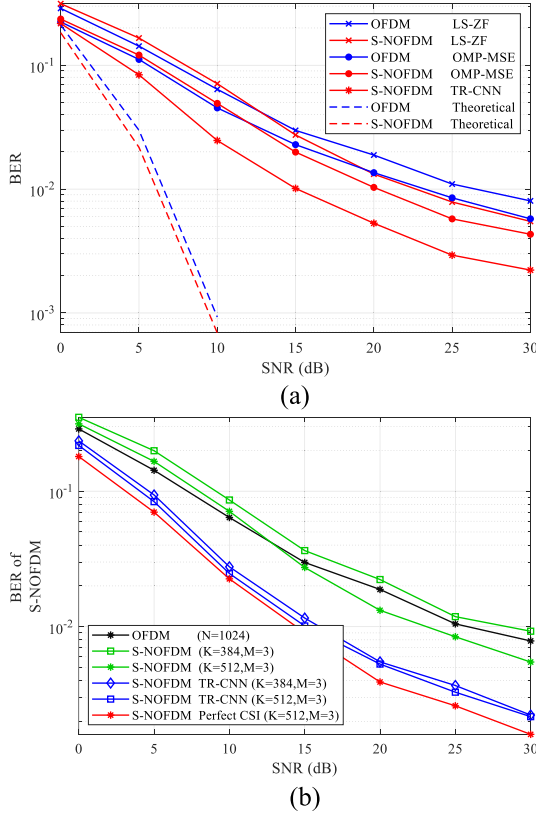


Fig. 6. BER performance comparisons of the proposed TR-CNN-based SNOFDM scheme and traditional method under UWA channel Env1. (a) BER with different channel estimation schemes. (b) BER with different subcarrier expansion.

the compressed sensing (CS) theory and can effectively take advantage of the sparse characteristic to estimate the UWA channel. In comparison to the TR-CNN receiver, i.e., OMP-MSE receiver, the TR-CNN receiver shows lower BER results. This indicates that the proposed TR-CNN receiver can better handle the complex UWA channel. In the proposed scheme, the T-R implementation in the network can reduce the multipath effect, whereas the TR-CNN can deal with ICI caused by nonorthogonality and Doppler shifts and delay spread of the UWA channel to recover the signals with lower BER.

However, there is still a big gap between the results of the proposed scheme and the ideal AWGN scenario. We believe that several factors including the complex UWA channel (Bellhop channel model used in the simulation), the lack of channel coding, and error correction code are the main reason contributing to this. For one thing, we focus on the TR-CNN receiver improvements by comparing it with other N-OFDM receivers, the channel coding and error correction code are not utilized to further improve the BER performance. For another, serious ICI and Doppler shift are introduced in UWA scenario simulations and the channel impact cannot be fully removed, thus leaving a big gap.

To demonstrate how the proposed scheme performs under the influence of a complex UWA scenario, we perform simulations under perfect knowledge of CSI and compare BER results with

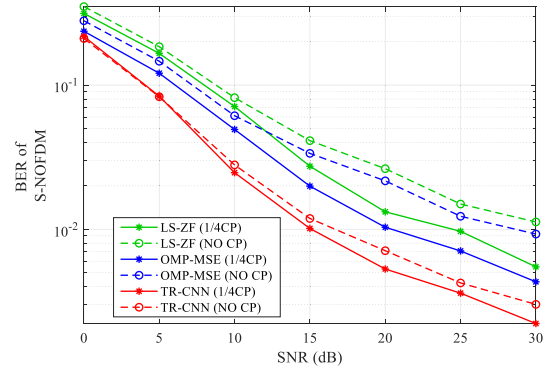


Fig. 7. BER performance comparisons of the proposed TR-CNN-based S-NOFDM scheme with different CP lengths.

other schemes in Fig. 6(b). The BER result under perfect knowledge of CSI shows an optimal result of the BER in the provided UWA channel. The BER under perfect knowledge of CSI is close to the BER of the TR-CNN receiver. This demonstrates that the TR-CNN successfully learns the characteristics of UWA channels and reduces the influence of the complex channel. High signal detection performance can be achieved by introducing TR-CNN in S-NOFDM systems.

In Fig. 6(b), we also investigated the influence of different subcarrier expanding factors M . Regarding the spectral efficiency, it is obvious that S-NOFDM has higher spectral efficiency and is $N/K = 2$ ($K = 512$), $N/K = 2.67$ ($K = 384$) times better than OFDM, which indicates that the S-NOFDM modulation scheme is attractive in UWA communication systems. However, the smaller the selected subcarriers K is, the larger the interference between SNOFDM symbols/subcarriers will be when the number of expanding subcarriers M is fixed. It implies that the spectral efficiency improvement of S-NOFDM is achieved at the cost of signal detection performance degradation, but the proposed TR-CNN-based S-NOFDM scheme is superior to the LS-ZF/OMP-MSE method for both OFDM modulation and NOFDM modulation. Our following analysis uses S-NOFDM only with $K = 512$, $M = 3$.

C. Robustness Analysis

1) *Impact of CP Length:* In Fig. 7, we analyze the signal performance of the proposed scheme with different CP lengths. We can see that for the LS-ZF and OMP-MSE methods, the reduction of the CP length leads to serious performance degradation due to the multipath effect of the UWA channel, and the gap becomes larger as the SNR increases. The proposed TR-CNN-based S-NOFDM scheme is insensitive to the CP length which is because we utilize the multipath information component to mitigate the long delay spread and fully learn the nonorthogonal correlation between subcarriers through the Conv Layers. It has strong robustness for different CP lengths, so the CP reduction or removal does not cause a large performance gap, which can make the TR-CNN-based S-NOFDM UWA communication system further improve the spectrum.

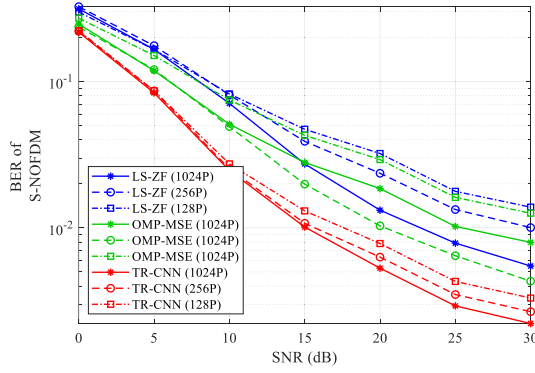


Fig. 8. BER performance comparisons of the proposed TR-CNN-based S-NOFDM scheme with different pilot numbers.

2) *Impact of Pilot*: Fig. 8 shows the robustness of the proposed scheme for different numbers of pilots, we can see that the proposed TR-CNN-based S-NOFDM scheme is robust for the number of pilots and the performance is superior to the traditional scheme. At low SNR, the ZF channel estimator itself cannot perform accurate channel estimation, and the BER performance is determined by the network's learning and generalization capability. At high SNR, reducing the number of pilots will make the estimated CIR less accurate, but it is only a slight degradation, and this performance loss is worthwhile to improve the spectrum utilization.

3) *Impact of the Mismatched Channel Environment*: To explore the robustness of the proposed TR-CNN-based S-NOFDM scheme for the mismatched channel environment, we generate three channels with different characteristics, and the parameters are given in Section IV-A specifically, Env2 introduces the random Doppler shifts caused by the relative movement of the transceiver compared to Env1, and Env3 increases both the maximum delay spread and the Doppler shifts, which will lead to an increasingly complex channel environment. To enhance the robustness of the network to different UWA channel environments, we fed two data streams consisting of estimated CIR and distorted received signals into the TR-CNN and mitigate the long delay spread through the convolution operation, while the Conv Layers of the TR-CNN fully exploit the inherent information between S-NOFDM symbols, which enables the pretrained network to be directly employed for testing various channel environments without additional training overhead.

The BER curves of the proposed TR-CNN-based S-NOFDM receiver, the LS-ZF receiver, and the OMP-MSE receiver under different channel environments are shown in Fig. 9. The LS-ZF receiver and OMP-MSE receiver are heavily dependent on the current channel condition and its signal detection performance degrades severely as the channel environment becomes harsh. In contrast, the inputs of the TR-CNN are the estimated CIR and the received information signal, the multipath component is used to mitigate the long delay spread, and the inherent information between S-NOFDM symbols is effectively extracted simultaneously through the Conv Layers, the TR-CNN can

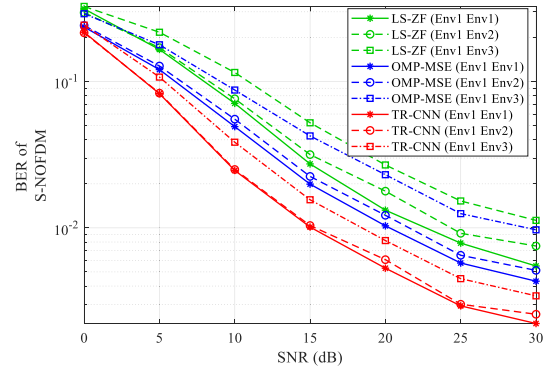


Fig. 9. BER performance comparisons of the proposed TR-CNN-based S-NOFDM scheme under different UWA channel environments.

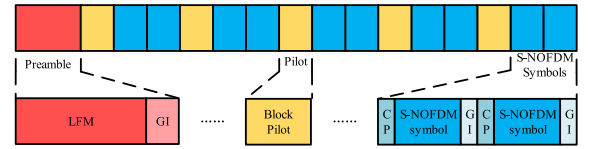


Fig. 10. Frame structure of S-NOFDM packet.

automatically learn the inherent feature information among S-NOFDM symbols without being limited by the current channel environment. This means that the proposed TR-CNN-based S-NOFDM scheme can be directly used in different conditions of the UWA channel environment. As shown in Fig. 9, the BER performance under Env2 almost does not increase, and under Env3 the BER performance slightly decreases but still outperforms the traditional ZF method. This demonstrates that the TR-CNN can tolerate a certain degree of Doppler shifts, and the TR-CNN-based S-NOFDM scheme is suitable for UWA channels with ISI/ICI, the proposed TR-CNN-based S-NOFDM scheme in the UWA communication system can achieve higher gain.

D. Experimental Results Analysis

To verify the performance of the proposed TR-CNN-based S-NOFDM scheme in a realistic UWA channel environment, we conducted a UWA communication experiment in the Haihe River, Tianjin, China, in September 2020.

In the river trial, the experimental system parameters are the same as the simulation parameters. At the transmitter, binary bits are processed and modulated in a mini-PC with S-NOFDM modulation algorithm installed. The packet size of each frame is 10240 random bits, and 80 packets are sent for each environment. The frame structure is illustrated in Fig. 10. Each packet contains 10 S-NOFDM symbols, and each symbol contains 1024 binary bits. The frame structure of the transmitted signal consists of linear frequency modulation (LFM), block pilot, and S-NOFDM symbols. The frame header LFM signal is used for synchronization and the block pilot signal is used for channel estimation to obtain the actual channel.

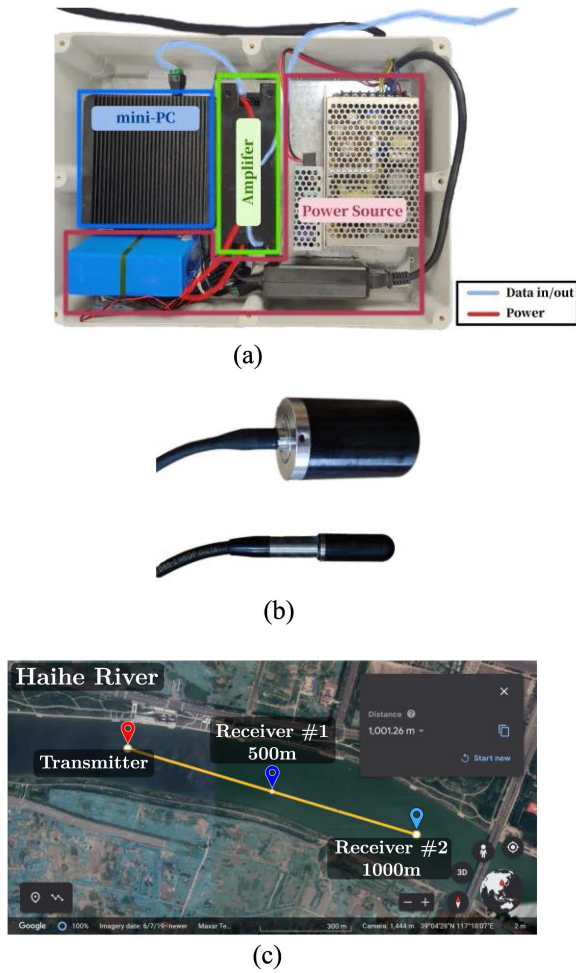


Fig. 11. Experimental equipment and field. (a) Transmitter and receiver devices. (The mini-PC is with build-in DAC/ADC and S-NOFDM modulation/demodulation algorithm installed.) (b) Transducers in experiments. (Top, WBT22-1601 transducer for transmission. Bottom, Brüel&Kjær Hydrophone Types 8104 for receiving), (c) Transmitter and receiver placement in Haihe River.

The carrier frequency is 25 kHz, the bandwidth is 20 to 30 kHz, and the sampling rate is 96 kHz. The constructed frame sequences are generated in the mini-PC and then passed to digital analog converter (DAC), to an amplifier, and finally to the transducer for transmission. The amplifier gain is changed with measured noise power to achieve multiple SNR tests. At the receiver, the signal is received with a transducer, later pass to an amplifier, then to the analog digital converter (ADC) and the mini-PC with the proposed TR-CNN S-NOFDM demodulation algorithm installed. The equipment is shown in Fig. 11(a) and (b). The transmitter and receiver devices are identical except for the installed algorithm and the amplifier in/out direction.

The depth at the experimental site is about 8 m, the depth of the transmitting transducer is about 1.5 m, and the depth of the receiving hydrophone is about 1.5 m. During the experiment, both the transmitter and receiver hydrophones are carried out using ships, which drift on the river surface. The experiment communication range was about 500 and 1000 m, as depicted in Fig. 11(c).

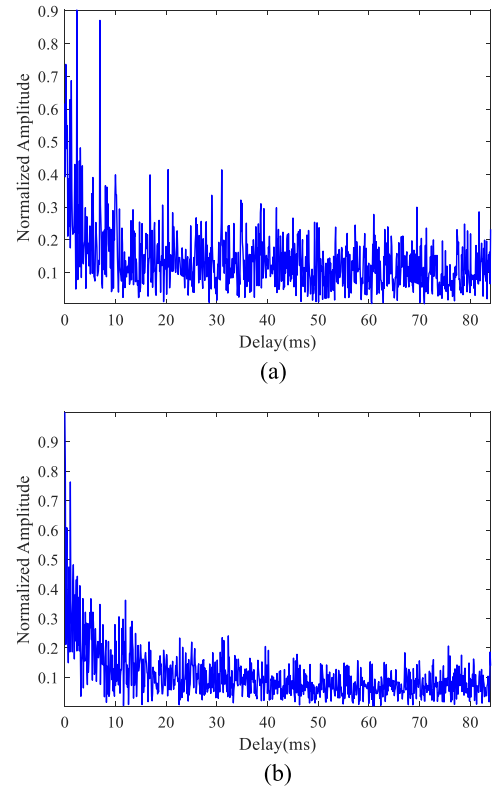


Fig. 12. Example of estimated CIR in a single transmission. (a) Estimated CIR: distance between the transmitter and receiver is 500 m. (b) Estimated CIR: distance between the transmitter and receiver is 1000 m.

To measure CIR in the lake trail, several transmissions during the experiment to obtain different moments of channel data, and we utilize the LS method to measure the channel environment for the communication experiment. Fig. 12(a) and (b) show the CIR at the current moment for 500 m and 1000 m, respectively, and it can be seen that the CIR of 500 m environment is more complex.

We analyze the robustness of the proposed TR-CNN-based S-NOFDM scheme through experiments. We pretrained the network with the simulated channel environment and use it directly for the UWA communication experiment. In each SNR test, the noise power is measured before transmission, and we changed the amplifier in the transmitter to achieve the target SNR. The same 80 packets are transmitted in different SNRs. Fig. 13 shows the performance of the proposed TR-CNN-based S-NOFDM receiver and the ZF-LS, OMP-MSE method in the experiment. Due to the long delay spread and high noise in shallow water environments and the harsh channel environment, the BER curve tends to be saturated with the increase of SNR in the ZF-LF and OMP-MSE method while the TR-CNN receiver shows much lower BER. Both in 500m and 1000 m environments, the results at high SNR reached 10^{-2} and is close to the optimal BER (Perfect CSI) results shown in Fig. 6(b). The proposed method demonstrates that the TR-CNN-based S-NOFDM scheme is promising to achieve reliable UWA communication, with less sensitivity to various channel conditions and high detection performance compared to the traditional LS-ZF or OMP-MSE method. It implies that the proposed TR-CNN-based S-NOFDM

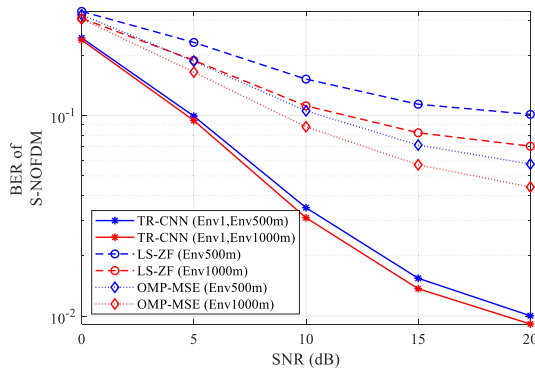


Fig. 13. BER performance comparisons of the proposed TR-CNN-based S-NOFDM scheme under experiment environment.

scheme once after training, can be directly used for the experiment and can adapt to the fast time-varying UWA channel environment without additional real-time training before transmission.

V. CONCLUSION

In this article, we present a novel T-R deep CNN-based S-NOFDM scheme for UWA communication. It can efficiently deal with the multipath information of the UWA channel and the interference between S-NOFDM symbols/subcarriers. To enhance the robustness and generalization of the network, we first utilize the multipath component together with CNN to simultaneously implement the T-R process and extract signal features. We consider the S-NOFDM complex signals as 2-D matrices to fully exploit the inherent information between S-NOFDM symbols. The multipath component is then used to mitigate the long delay spread and the inherent information between S-NOFDM symbols is effectively extracted simultaneously through the convolutional layer. Simulations and experiments validate that the proposed TR-CNN-based S-NOFDM scheme can achieve higher detection performance compared to the traditional method. Also, the proposed TR-CNN-based S-NOFDM scheme shows robustness for various channel environments, different CP lengths, and the number of pilots.

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